

BTEC Level 3 Applied Science • Unit 1 Physics • Worksheet

Wave Properties

Original JDSience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: wave type, phase, superposition and polarisation.

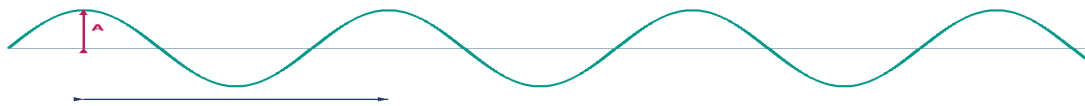
A Retrieval

1. Define a transverse wave and a longitudinal wave.

2. Give one example of each.

3. State the principle of superposition.

Transverse wave



Oscillation is perpendicular to the direction of energy transfer.

Longitudinal wave

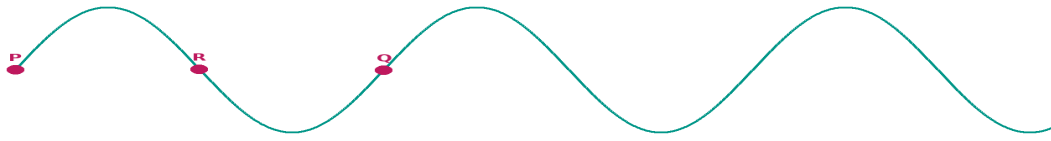


Oscillation is parallel to energy transfer. Particles bunch (compression) then spread (rarefaction).

Sound in air is longitudinal. Light is transverse. A sine-graph of sound is a graph, not a picture of the particles.

4. What is a compression and what is a rarefaction?

Phase: points a whole number of wavelengths apart



P and Q are in phase (separated by 1.0λ). P and R are in antiphase (0.5λ).

5. When are two points on a wave in phase?

6. Why can light be polarised but sound in air cannot?

7. Classify ultrasound, microwaves and stadium sound as transverse or longitudinal.

8. Two crests are 6.0 cm apart and $\lambda = 2.0$ cm. Are they in phase? Explain.

Working

D Exam-style practice

9. Describe the difference between transverse and longitudinal waves, with an example of each. [4]

10. State the principle of superposition. [2]

11. Explain why sound from a loudspeaker cannot be polarised. [2]

E Challenge

12. A student says a sine-wave drawing of sound proves sound is transverse. Write a correction a teacher could use.
